Six math clues · answer key

How a Sidewalk Is Made and How It Works

Grade 2 math. Join the amounts for addition or find what remains for subtraction. Each story takes one step.

Math 1 · 65

36 + 29 = 65 stones. Add 30 to get 66, then take away 1.

Math 2 · 35

82 − 47 = 35 squares. Subtract 40 to get 42, then subtract 7.

Math 3 · 83

45 + 38 = 83 pebbles. Add 40 to get 85, then subtract 2.

Math 4 · 43

71 − 28 = 43 pebbles. Subtract 30 to get 41, then add 2.

Math 5 · 28

7 + 21 = 28 days. Add 20 to get 27, then add 1.

Math 6 · 24

72 − 48 = 24 grooves. Subtract 50 to get 22, then add 2.

Grade 2 math. Six one-step addition and subtraction stories stay within 100. Quantities are supplied in the questions; there is no need to count painted details.

Reading & writing · answer key

How a Sidewalk Is Made and How It Works

Reading · 1

Cement is the gray powder, but concrete also needs water, sand and gravel. Cement and water make a paste that coats the sand and stones and holds them together.

Reading · 2

Concrete hardens through a chemical reaction between cement and water, not just by losing water. Curing keeps the slab damp so the reaction can continue and the slab gets stronger.

Reading · 3

A groove makes a thin line, so a crack tends to follow it neatly. Packed gravel helps support the slab evenly, so the weight spreads out and the slab bends less.

Writing

Accept a spoken, dictated, drawn, or written sentence that tells one step of how concrete is made or why a groove helps.

Cement is a fine gray powder that makes a paste with water, and it is only roughly 10 to 15 percent of the mix by volume. Concrete is cement plus sand, gravel and water. The cement and water react chemically (hydration), which is why concrete hardens even under water, and the slab is kept damp for about a week (curing) while most of its strength builds in the first month. Concrete shrinks slightly as it hardens, and grooves (control joints) give cracks a planned, neat line. Concrete is strong when squeezed and weak when bent, and a compacted gravel base helps support a slab evenly. Safety: wet concrete and dry cement powder can burn skin and irritate eyes and lungs, so children stay well back and never touch or taste either. The table-top activities use only flour, rice, chalk and crackers; check for food allergies, and keep the flour mix out of mouths. Chalk can snap into small pieces, so keep them off the floor.

Offer dictation: let the child say a complete thought while you write it, then invite them to add a picture, label, or sentence.

Four concrete clues

How a Sidewalk Is Made and How It Works

Quiz 1. What is cement?

Cement is a fine gray powder. With water it makes the paste that holds sand and stones together.

Quiz 2. How does wet concrete turn hard?

Cement and water react and link together, so concrete hardens even under water.

Quiz 3. Why is a new slab kept damp for days?

Curing keeps the slab damp so the hardening reaction can continue.

Quiz 4. What is a groove in a sidewalk for?

A groove makes a thin line, so a crack tends to follow it instead of wandering.

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